FedPASA: Reliability-Aware Semantic Adaptation for Cross-Domain Federated Recommendation

Anonymous Author(s)

Abstract

Federated recommendation (FR) enables collaborative model training while keeping user interactions local. However, transferring recommendation knowledge across services remains challenging due to disjoint item catalogs and uneven target-domain supervision. Frequently observed items provide abundant interaction evidence, whereas low-frequency items offer limited supervision, and client-specific preferences further introduce heterogeneity into semantic adaptation. Existing approaches commonly employ shared transformations or client-level personalization without explicitly accounting for differences in item-level supervision. To address these challenges, we propose FedPASA, a reliability-aware semantic adaptation framework for cross-domain federated recommendation. FedPASA introduces Popularity-Gated Dual-path Routing (PGDR), which combines two residual adaptation paths using item-frequency-derived weights. A private client profile further conditions the adapted representations through feature-wise scaling and shifting, while AutoAnchor incorporates pretrained semantic information through catalog-level anchoring. Extensive experiments on real-world cross-domain recommendation benchmarks demonstrate the effectiveness of FedPASA. The proposed framework achieves superior average cross-domain recommendation performance over competitive baselines under both sampled and full-catalog ranking. Further analyses demonstrate that frequency-aware routing improves recommendation accuracy beyond additional adapter capacity, while client conditioning and semantic anchoring provide complementary adaptation capabilities.

CCS Concepts

• **Information systems** → **Recommender systems**; • **Computing methodologies** → *Distributed computing methodologies.*

Keywords

Federated Recommendation; Cross-Domain Recommendation; Semantic Transfer; Long-tail Recommendation

ACM Reference Format:

Anonymous Author(s). 2027. FedPASA: Reliability-Aware Semantic Adaptation for Cross-Domain Federated Recommendation. In *Proceedings of the ACM Web Conference 2027 (WWW '27).* ACM, New York, NY, USA, 12 pages.

1 Introduction

With the growth of e-commerce and digital services, recommender systems have become essential infrastructure for online platforms. However, traditional recommender systems typically consolidate vast amounts of user data centrally, leading to privacy and regulatory concerns. Federated learning allows multiple clients to collaboratively train a model without sharing raw data; federated recommendation (FR) applies this idea to recommendation, where interaction histories remain on local devices and only model parameters are aggregated [1, 19].

Meanwhile, cross-domain transfer is another longstanding need: platforms wish to reuse a mature catalog to bootstrap a new service, where newly listed items lack behavioral evidence. However, existing cross-domain recommenders typically assume overlapping users or alignable item identifiers, and neither premise holds under federated constraints—participating services cannot pool user histories, and disjoint catalogs share no identifiers to align. We study this disjoint-catalog setting without requiring linked users between source and target domains.

A viable route is offered by pretrained text encoders: items from different catalogs are mapped into a common semantic space, enabling transfer without matching identifiers [29]. However, the semantic space alone does not resolve the uneven supervision of the target domain. On the one hand, frequently observed items supply many ranking constraints, whereas rare items provide little evidence for adaptation; on the other hand, clients observe different semantic regions through their own histories, so a globally rare item may be familiar to one client yet unseen by another. Moreover, excessive adaptation drifts representations away from the pretrained semantics, harming items with scarce evidence. An adaptation rule therefore needs to account for the amount of item-level evidence, the client context in which it is observed, and how much of the pretrained prior should be preserved.

Personalized adapters and expert mixtures address variation across clients [14, 27], but client-level personalization does not by itself specify how items with different training frequencies should share adaptation paths: a single shared transformation applies the same parameterization to all items, and an additional adapter introduces capacity without determining how to allocate it. In this light, we use item frequency as an observable proxy for the available target supervision, route each item between two residual paths, and adjust the routed representation to the local interaction profile through client conditioning.

To address the challenges mentioned above, we propose FedPASA, which combines three operations in a shared semantic adapter. Specifically, Popularity-Gated Dual-path Routing (PGDR) mixes a source-initialized path and a target-initialized path using a frequency-derived weight; client conditioning produces feature-wise scales and shifts from a locally pooled semantic profile; in addition, AutoAnchor interpolates the resulting representation with the frozen text embedding, with the anchor strength set by a catalog-level user-to-item ratio. Both residual paths are updated on the target domain, while source initialization provides a starting point rather than a fixed representation.

We evaluate this design on multiple real-world datasets, with controls that separate frequency assignment, model capacity, and final personalization. Our contributions are summarized as follows:

WWW '27, Dublin, Ireland
2027.

- We propose FedPASA, a reliability-aware semantic adaptation framework that adjusts item representations according to evidence strength at the item, client, and catalog levels. Our framework is motivated by an analysis of the failure of uniform adaptation in existing federated semantic transfer methods under uneven target supervision: a single shared transformation underfits well-supported frequent items while overfitting sparsely observed items. The proposed framework applies to different backbone encoders.
- We design PGDR to combine two trainable residual adaptation paths using frequency-derived weights, without an additional learned routing network. Client conditioning adjusts the combined representations using local semantic profiles, while AutoAnchor controls their mixture with the frozen semantic prior through a catalog-level rule.
- We conduct extensive experiments on multiple real-world datasets. With the same text encoder, FedPASA achieves higher average cross-domain recommendation accuracy than the evaluated baselines. Comparisons that retain the same network structure and parameter count further show that frequency-derived mixing weights outperform using a common pair of weights for all items or shuffling the item–weight assignments. Ablation studies further characterize the effects of individual components.

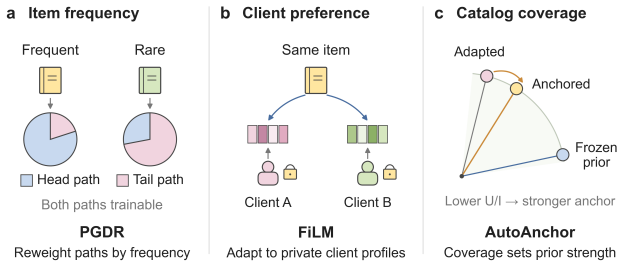


Figure 1: Three adaptation controls: (a) frequency-based path mixing; (b) private-profile conditioning; (c) catalog-dependent prior preservation.

2 Related Work

2.1 Federated Semantic Transfer

FR learns shared parameters from decentralized interactions [1, 19]; proximal optimization and local components address heterogeneity [2, 5, 10, 11]. Transferring item-ID models across disjoint catalogs requires alignment or content. TransFR combines pretrained language representations, adapter tuning, and local personalization [29]. FedDCSR disentangles shared and domain-specific sequential factors [30]; PFCR and FFMSR transfer text-derived content [6, 16]; FedCRF uses semantic graphs and contrastive fusion [7]. FeDecider personalizes LoRA aggregation with client-specific directional weights [8]. Beyond sequence modeling, fusion, and aggregation, FedPASA studies item-level supervision allocation within a shared semantic adapter.

2.2 Personalization and Preservation

FedPA combines private user adapters, shared group adapters, and adaptive gating [27]; FedVLR personalizes multimodal expert mixtures [14]. ESAM addresses long-tail imbalance [4], MeLU adapts user preferences [9], and multi-gate MoE learns input-dependent expert combinations [17]. PGDR instead derives deterministic routing weights from target item frequency. Learned, constant, and shuffled routing provide comparisons that isolate this allocation rule.

FiLM applies feature-wise scales and shifts [22]; FedPASA generates them from a private client profile within a shared adapter. L2-SP and DELTA preserve source information through parameter or feature penalties [12, 13]. AutoAnchor instead interpolates the normalized ranking representation with the frozen semantic prior. Its fixed catalog-level coverage heuristic is motivated by bias–variance and geometric analyses and assessed through ranking experiments.

3 FedPASA

3.1 Problem Setup and Overview

We consider cross-domain federated recommendation with disjoint item catalogs. Let $\mathcal{C}$, $\mathcal{U}$, and $\mathcal{I}$ denote the sets of clients, users, and target items, respectively. Each client c maintains a subset of users $\mathcal{U}_c$ and their interaction histories locally. A frozen text encoder maps each item i to a unit-normalized semantic representation $x_i \in \mathbb{R}^d$, providing a common representation space across domains. Each user u is associated with a private embedding $e_u \in \mathbb{R}^d$ that remains on its corresponding client.

FedPASA adapts the frozen semantic representations through three mechanisms: Popularity-Gated Dual-path Routing (PGDR), Private-Profile Client Conditioning, and Catalog-Level Semantic Anchoring (AutoAnchor). Source-domain training initializes one residual adaptation path, while the second path is randomly initialized on the target domain. Both paths remain trainable during target-domain federated optimization.

For user u on client c, the recommendation score is defined as

$$s(u,i) = e_u^\top z_{c,i}^{\text{anchor}}. \tag{1}$$

The model is optimized using Bayesian Personalized Ranking (BPR) [24]:

$$\mathcal{L}_{\text{BPR}} = -\mathbb{E}_{(u,i,j)} \log \sigma\big(\tau[s(u,i) - s(u,j)]\big),$$

where i is an observed positive item, j a sampled negative item, and τ controls the score scale.

Figure 2 illustrates the overall FedPASA framework, and Table 11 summarizes the notation. The forward computation consists of three consecutive stages:

$$x_i \xrightarrow{\text{routing}} h_i \xrightarrow[\text{and residual}]{\text{FiLM}} z_{c,i} \xrightarrow{\text{anchoring}} z_{c,i}^{\text{anchor}}. \tag{2}$$

3.2 Popularity-Gated Dual-path Routing

Target items exhibit substantially different interaction frequencies, resulting in different amounts of supervision for semantic adaptation. PGDR explicitly uses item frequency to determine the relative contributions of two residual adaptation paths, enabling frequency-dependent representation learning within a shared model.

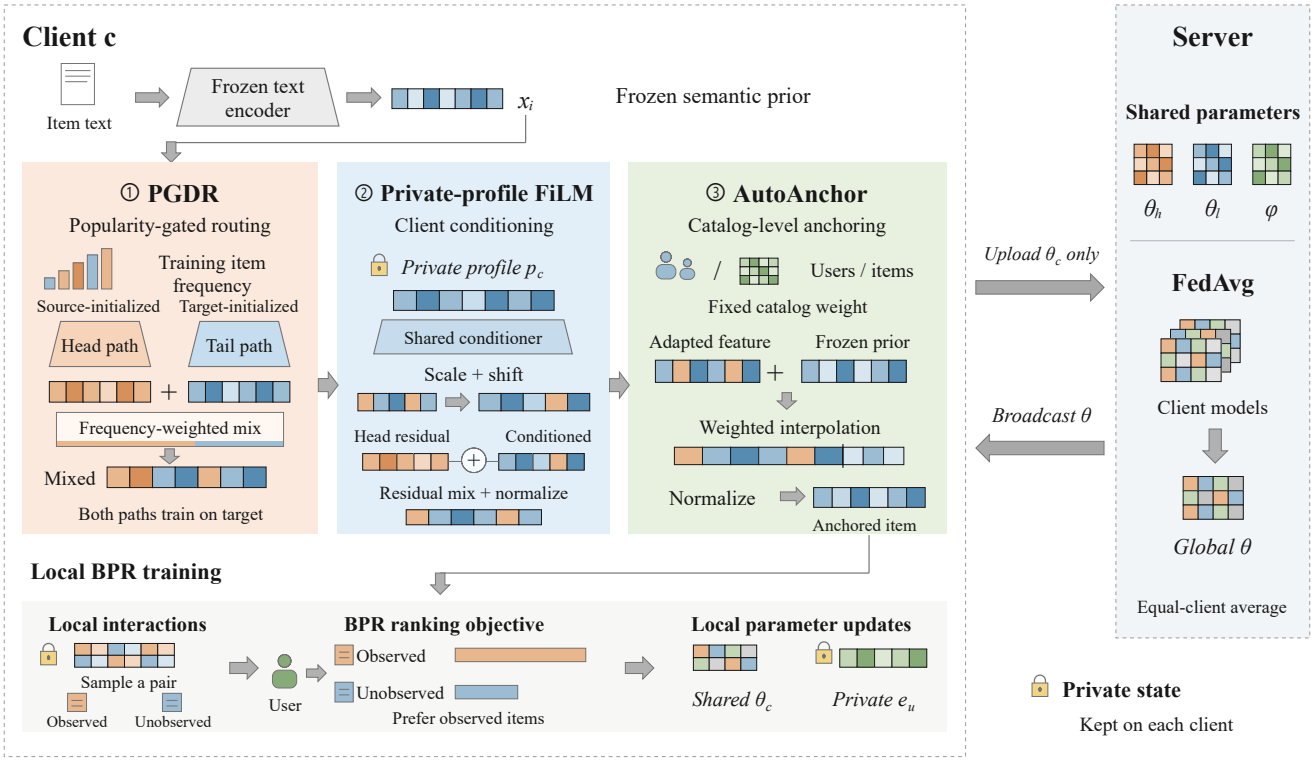


Figure 2: FedPASA framework. PGDR combines source-initialized and target-initialized residual adapters according to item frequency. Private-profile FiLM conditions the resulting representations on local interaction semantics, while AutoAnchor integrates the frozen semantic prior. Clients optimize shared model parameters and private user embeddings using BPR, and the server aggregates the shared parameters through FedAvg.

Let $\mathrm{freq}(i)$ denote the number of target-domain training interactions containing item i. We first construct an inverse-frequency score $\bar{q}_i = 1/\sqrt{\mathrm{freq}(i)+1}$ and normalize it over the target catalog:

$$q_i = \frac{\bar{q}_i - \min_j \bar{q}_j}{\max_j \bar{q}_j - \min_j \bar{q}_j + \epsilon}. \tag{3}$$

A threshold t and exponent κ further reshape the routing score:

$$q'_i = \mathrm{clip}\left(\tfrac{q_i - t}{1-t}, 0, 1\right)^{\kappa}.$$

The resulting score q'_i increases as item frequency decreases and controls the relative contribution of the target-initialized path.

PGDR contains two residual adapters, denoted by A_h and A_l. The head adapter A_h is initialized from the source-domain adapter, providing transferred semantic adaptation knowledge. The tail adapter A_l is randomly initialized on the target domain and learns complementary transformations from target interactions. Each adapter contains two sequential residual blocks, $v \leftarrow v + F_\ell(v)$, where F_ℓ is a bottleneck multilayer perceptron, followed by ℓ_2 normalization. Both adapters are optimized during target-domain federated training.

The routed item representation is

$$h_i = (1 - q'_i)\, A_h(x_i) + q'_i\, A_l(x_i). \tag{4}$$

Frequently observed items place greater weight on the source-initialized head path, whereas lower-frequency items receive a larger contribution from the target-initialized tail path. The mixed representation h_i is subsequently passed to the client-conditioning module.

3.3 Private-Profile Client Conditioning

Item frequency characterizes supervision at the item level, while clients may observe substantially different semantic regions of the target catalog. To incorporate this client-side heterogeneity, FedPASA constructs a private semantic profile from each client's local interaction history and uses it to condition the routed representation.

For client c, let $\mathcal{I}_c$ denote the multiset of items appearing in its training interactions. Repeated interactions retain their multiplicity in the profile construction. The client profile is defined as

$$p_c = \mathrm{norm}\left(\tfrac{1}{|\mathcal{I}_c|} \sum_{i \in \mathcal{I}_c} x_i\right). \tag{5}$$

A shared bottleneck network $g_\phi : \mathbb{R}^d \to \mathbb{R}^{2d}$ maps the profile into feature-wise FiLM parameters:

$$(\gamma_c, \beta_c) = g_\phi(p_c). \tag{6}$$

The routed representation h_i is then modulated as

$$m_{c,i} = h_i \odot (1 + \lambda \tanh \gamma_c) + \lambda \beta_c,$$
$$z_{c,i} = \mathrm{norm}\big((1-\alpha)A_h(x_i) + \alpha m_{c,i}\big). \tag{7}$$

The coefficient λ controls the strength of client-dependent modulation. The tanh transformation bounds the multiplicative factor to $[1-\lambda, 1+\lambda]$, while the additive term provides learned client-specific feature shifts. The coefficient $\alpha \in [0,1]$ controls the contribution of the conditioned representation relative to the head-path residual.

For the conditioning module, only the shared parameters ϕ participate in federated communication. The client profile p_c is computed from local training interactions and remains on the client.

3.4 Catalog-Level Semantic Anchoring

After frequency-aware routing and client conditioning, FedPASA further integrates the frozen semantic representation into the final ranking representation. AutoAnchor determines the contribution of this semantic prior according to target-catalog coverage.

We characterize target-catalog coverage using the user-to-item ratio

$$\eta_T = |\mathcal{U}_T|/|\mathcal{I}_T|,$$

where $|\mathcal{U}_T|$ denotes the number of observed target-domain users and $|\mathcal{I}_T|$ denotes the target catalog size. Given coverage thresholds $\tau_l < \tau_h$ and maximum anchor weight ρ, AutoAnchor computes

$$\rho_{\text{eff}} = \rho \cdot \mathrm{clip}\Big(\frac{\tau_h - \eta_T}{\tau_h - \tau_l}, 0, 1\Big),$$
$$z^{\text{anchor}}_{c,i} = \mathrm{norm}\big((1-\rho_{\text{eff}})\, z_{c,i} + \rho_{\text{eff}}\, x_i\big). \tag{8}$$

The effective weight ρ_{eff} controls the contribution of the frozen semantic prior. When $\eta_T \le \tau_l$, the maximum anchor weight ρ is used. As η_T increases between τ_l and τ_h, the prior contribution decreases linearly. For $\eta_T \ge \tau_h$, the final representation relies directly on the adapted representation. Its weight is determined from target-domain catalog statistics and remains fixed throughout federated training and evaluation.

The default catalog-level gate introduces no additional trainable parameters. We additionally evaluate a per-item learnable alternative,

$$\rho_i = \rho\, \sigma(a_i),$$

where $a \in \mathbb{R}^{|\mathcal{I}|}$ contains item-specific logits included in federated aggregation. The corresponding comparison is reported in the anchor analysis (§4.5.1).

3.5 Federated Training and Local Personalization

FedPASA jointly learns shared semantic adaptation parameters across clients while maintaining private user representations locally. The shared trainable parameters are $\theta = \{\theta_h, \theta_l, \phi\}$, where θ_h and θ_l denote the parameters of the two residual adapters and ϕ denotes the parameters of the client-conditioning network. Each client c additionally maintains a private user-embedding table $E_c = \{e_u : u \in \mathcal{U}_c\}$.

During communication round r, the server broadcasts the current shared parameters $\theta^{(r)}$ to all participating clients. Each client initializes its local shared-model copy from $\theta^{(r)}$, retains its private user embeddings from the preceding round, and performs K local optimization steps using the BPR objective. The local optimization updates both the shared parameters θ_c and private user embeddings E_c. After local training, each client uploads only θ_c to the server. Under the default full-participation setting, the server performs equal-client FedAvg:

$$\theta^{(r+1)} = \frac{1}{|C|} \sum_{c \in C} \theta_c.$$

The federated adaptation stage optimizes the BPR objective without an additional proximal penalty.

After R communication rounds, the resulting global parameters $\theta^{(R)}$ define the shared FedPASA model. We refer to this model before final local personalization as *FedPASA base*. For final personalization, each client initializes its local model from $\theta^{(R)}$, retains its learned private user embeddings, and optimizes

$$\mathcal{L}_{\text{pers},c} = \mathcal{L}_{\text{BPR},c} + \frac{\mu_p}{2} \|\theta_c - \theta^{(R)}\|_2^2.$$

The proximal term encourages local adaptation around the final global solution [10, 11]. During this stage, both θ_c and E_c are updated, while the proximal penalty is applied only to the shared-model parameters θ_c. Each client performs K_p local personalization steps. The resulting personalized parameters remain local and are not uploaded to the server.

Algorithm 1 summarizes the complete training procedure. Concrete optimization settings are provided in the experimental setup (§4.2).

Initialization and local optimization. Source-domain training uses one residual adapter together with source-domain user embeddings. The resulting adapter parameters initialize the target-domain head path. The tail path is randomly initialized. The final linear layer of the conditioning network is zero-initialized so that client modulation starts from the shared representation.

Target item frequencies are computed exclusively from target training interactions. Each client profile is computed from its own

Algorithm 1 FedPASA training and local personalization

Require: Frozen item features, training histories, clients C, rounds R, local steps K, personalization steps K_p, proximal coefficient μ_p

For cross-domain training, optimize a source-domain residual adapter by source BPR and equal-client FedAvg
Initialize θ_h from the source adapter; randomly initialize θ_l and ϕ
Compute target routing scores q_i from target training frequencies
Compute each client profile p_c from its training interactions
Compute and fix the catalog-level anchor weight ρ_{eff}
Initialize private user tables $E_c = \{e_u : u \in \mathcal{U}_c\}$ for all clients
$\theta^{(0)} \leftarrow \{\theta_h, \theta_l, \phi\}$
for $r = 0, \ldots, R-1$ **do**
  Broadcast $\theta^{(r)}$ to all participating clients
  for all $c \in C$ **in parallel do**
    $\theta_c \leftarrow \theta^{(r)}$; retain E_c from the preceding round
    Take K local steps on $\mathcal{L}_{\text{BPR}}$, updating (θ_c, E_c)
    Keep E_c local and upload θ_c
  end for
  $\theta^{(r+1)} \leftarrow |C|^{-1} \sum_{c \in C} \theta_c$
end for
Preserve $\theta^{(R)}$ as the FedPASA base model
for all $c \in C$ **do**
  $\theta_c \leftarrow \theta^{(R)}$; retain the final E_c
  $\ell_c \leftarrow \mathcal{L}_{\text{BPR}} + \frac{\mu_p}{2} \|\theta_c - \theta^{(R)}\|_2^2$
  Take K_p local personalization steps, updating (θ_c, E_c)
end for
return Personalized (θ_c, E_c) stored on each client

training-item multiset, with repeated interactions retained. Routing statistics, client profiles, and the catalog-level anchor weight are fixed throughout federated optimization, personalization, and evaluation.

Private user embeddings persist across communication rounds. AdamW optimizer states are reset at the beginning of each local optimization phase. During local training, users are sampled uniformly within a client and positive items are sampled uniformly from the corresponding user's training history. Negative items are sampled from items outside the user's observed training interactions. Dropout is enabled during model updates and disabled during evaluation.

The communication process exchanges only the shared adaptation and conditioning parameters. Raw interaction histories, client profiles, and private user embeddings remain on their respective clients.

3.6 Analysis of the Adaptation Controls

We analyze the main adaptation mechanisms from three perspectives: the relationship between target-estimation reliability and semantic-prior weighting, the geometry of normalized semantic anchoring, and the interaction between popularity routing and client conditioning.

3.6.1 Bias–Variance Tradeoff of Prior Weighting.

Proposition 1 (Bias–variance tradeoff). *Let z^* denote an ideal target representation and x a semantic prior. Consider an estimated target representation $\hat{z} = z^* + \varepsilon$, where $\mathbb{E}\varepsilon = 0$ and $\mathbb{E}\|\varepsilon\|_2^2 = \sigma^2/r$ with $r, \sigma^2 > 0$. Here, r characterizes the reliability of the target estimate: larger r corresponds to lower estimation variance. Let $b^2 = \|x - z^*\|_2^2$ denote the squared bias of the semantic prior. For the interpolated estimate $\tilde{z}_\omega = (1-\omega)\hat{z} + \omega x$, we have*

$$\mathbb{E}\|\tilde{z}_\omega - z^*\|_2^2 = (1-\omega)^2\sigma^2/r + \omega^2 b^2, \qquad \omega^* = \frac{\sigma^2}{\sigma^2 + rb^2}. \tag{9}$$

When $b > 0$, ω^* decreases as r increases.

The optimal weight increases with estimation noise and decreases with prior bias: when target evidence is limited, more of a useful prior should be retained. AutoAnchor operationalizes this principle at the catalog level by using target-catalog coverage to determine the contribution of the frozen semantic prior.

3.6.2 Geometry of Normalized Semantic Anchoring.

Proposition 2 (Normalized anchor geometry). *For fixed unit vectors x and z, define $w_\omega = (1-\omega)z + \omega x$, $y_\omega = w_\omega/\|w_\omega\|_2$, and $a = x^\top z$. For $0 \le \omega < 1/2$,*

$$\|w_\omega\|_2 \ge 1 - 2\omega > 0, \qquad \frac{d(x^\top y_\omega)}{d\omega} = \frac{(1-\omega)(1-a^2)}{\|w_\omega\|_2^3} \ge 0. \tag{10}$$

Therefore, $1 - x^\top y_\omega$ is non-increasing with respect to the anchor weight ω.

This proposition characterizes the geometric effect of normalized semantic interpolation: for a fixed adapted representation, increasing the anchor weight moves the final representation toward the frozen semantic prior. The experimental analysis (§4.5.1) further examines the representation geometry produced after federated optimization.

Table 1: Dataset statistics after the three-interaction filter. N includes training, validation, and test interactions.

Dataset	Users	Items	N
Music	12,315	10,680	339,923
Books	262,698	170,389	10,848,388
Movies	28,226	19,252	1,056,608

3.6.3 Client Conditioning.

The multiplicative part of the FiLM transformation in Eq. (7) is $1 + \lambda \tanh \gamma_c$. Since $\tanh \gamma_c \in [-1, 1]$, the resulting feature-wise multiplicative factors satisfy $1 - \lambda \le 1 + \lambda \tanh \gamma_c \le 1 + \lambda$. Thus, λ directly controls the range of client-dependent multiplicative modulation, while the learned additive term $\lambda\beta_c$ shifts the routed representation according to the client's semantic profile.

3.6.4 Interaction of Routing and Client Conditioning.

PGDR and client conditioning act on the same semantic representation but control different aspects of adaptation. To make the interaction explicit, consider the case in which FiLM modulation is disabled, i.e., $\lambda = 0$. Combining Eqs. (4) and (7) gives

$$z_{c,i} = \mathrm{norm}\big((1 - \alpha q_i')A_h(x_i) + \alpha q_i' A_l(x_i)\big). \tag{11}$$

The effective coefficient of the target-initialized path is therefore $\alpha q_i'$. The item-frequency-derived score q_i' determines how the two adaptation paths are allocated to each item, while α controls the overall strength of this allocation. Even for the largest routing score $q_i' = 1$, the head path retains a contribution of $1 - \alpha$.

When FiLM is enabled, client-dependent feature scales and shifts are applied to the frequency-routed representation before normalization and semantic anchoring. Consequently, item frequency controls path combination, whereas the private client profile controls feature-level adjustment.

4 Experiments

We evaluate FedPASA through four research questions:

- **RQ1:** How does FedPASA compare with federated recommendation and semantic-transfer baselines?
- **RQ2:** How much does each component contribute beyond additional adapter capacity?
- **RQ3:** How do item frequency, client exposure, and catalog coverage explain the gains?
- **RQ4:** Does FedPASA remain effective under client heterogeneity, restricted computation, and alternative encoders?

4.1 Datasets

We evaluate cross-domain recommendation on three Amazon datasets. The benchmark contains all six ordered transfers among Amazon Music (CDs & Vinyl), Books, and Movies (Movies & TV) [21]; their item catalogs are disjoint and no linked users are required. Table 1 summarizes the retained data. Users with fewer than three interactions are removed; the last interaction is used for testing, the preceding interaction for validation, and all earlier interactions for training. Appendix A gives the evaluation-user counts and preprocessing details.

Table 2: Sampled-ranking comparison on the first three transfers. FedPASA includes final local personalization. Bold and underline mark the best and second-best results; run counts are given in Appendix B.

Transfer	Music→Books		Music→Movies		Books→Movies	
Method	HR@20	NDCG@20	HR@20	NDCG@20	HR@20	NDCG@20
FeDecider [8]	0.2011	0.0714	<u>0.4278</u>	<u>0.1823</u>	<u>0.3981</u>	<u>0.1705</u>
FFMSR [16]	0.0273	0.0116	0.1354	0.0824	0.1231	0.0720
FedDCSR [30]	0.0254	0.0218	0.0924	0.0724	0.0924	0.0724
FedMF [1]	0.2005	0.0715	0.2127	0.0792	0.2127	0.0792
FedAvg [19]	0.1949	0.0672	0.3440	0.1388	0.3440	0.1388
FedProx [11]	0.1998	0.0705	0.3381	0.1351	0.3381	0.1351
FedAdam [23]	0.1986	0.0690	0.3398	0.1359	0.3398	0.1359
TransFR [29]	<u>0.2058</u>	<u>0.0723</u>	0.3292	0.1283	0.3330	0.1307
L2-SP [12]	0.2032	0.0718	0.3244	0.1244	0.3141	0.1229
FedPer [2]	0.2031	0.0714	0.3410	0.1353	0.3492	0.1397
pFedMe [5]	0.2047	0.0719	0.3284	0.1285	0.3368	0.1332
FedMoE [20]	0.2039	0.0707	0.3424	0.1355	0.3566	0.1444
FedPASA	**0.2719**	**0.1159**	**0.4980**	**0.2254**	**0.4948**	**0.2284**

Table 3: Sampled-ranking comparison on the remaining transfers and the six-transfer mean. Metrics and formatting follow Table 2.

Transfer	Books→Music		Movies→Books		Movies→Music		Six-transfer mean	
Method	HR@20	NDCG@20	HR@20	NDCG@20	HR@20	NDCG@20	HR@20	NDCG@20
FeDecider [8]	<u>0.5355</u>	<u>0.2283</u>	0.2058	<u>0.0734</u>	<u>0.5380</u>	<u>0.2284</u>	<u>0.3844</u>	<u>0.1590</u>
FFMSR [16]	0.1585	0.0841	0.0214	0.0094	0.1701	0.0992	0.1060	0.0598
FedDCSR [30]	0.1090	0.0832	0.0254	0.0218	0.1090	0.0832	0.0756	0.0591
FedMF [1]	0.2212	0.0826	0.2005	0.0715	0.2212	0.0826	0.2115	0.0778
FedAvg [19]	0.5074	0.2129	0.1949	0.0672	0.5074	0.2129	0.3488	0.1396
FedProx [11]	0.5024	0.2093	0.1998	0.0705	0.5024	0.2093	0.3468	0.1383
FedAdam [23]	0.5073	0.2098	0.1986	0.0690	0.5073	0.2098	0.3485	0.1382
TransFR [29]	0.5046	0.2121	0.2001	0.0705	0.5048	0.2115	0.3462	0.1376
L2-SP [12]	0.4996	0.2083	0.2017	0.0710	0.5040	0.2122	0.3412	0.1351
FedPer [2]	0.5052	0.2150	0.2040	0.0721	0.5027	0.2099	0.3509	0.1406
pFedMe [5]	0.5054	0.2131	0.2026	0.0712	0.5064	0.2117	0.3474	0.1383
FedMoE [20]	0.5088	0.2143	<u>0.2064</u>	0.0731	0.5107	0.2141	0.3548	0.1420
FedPASA	**0.5408**	**0.2403**	**0.2824**	**0.1249**	**0.5462**	**0.2439**	**0.4390**	**0.1965**

Table 4: Joint frequency–exposure ablation before local personalization under full-catalog ranking. C/U: low-frequency items locally seen/unseen; W: frequency > 5.

Model	C: low-frequency / seen		U: low-frequency / unseen		W: frequency > 5	
	HR@20	NDCG@20	HR@20	NDCG@20	HR@20	NDCG@20
FedPASA base	**0.0160**	**0.0070**	**0.0347**	**0.0151**	**0.0102**	**0.0037**
w/o PGDR	0.0041	0.0010	0.0039	0.0013	0.0023	0.0007
w/o FiLM	0.0041	0.0018	0.0276	0.0123	0.0037	0.0014
w/o AutoAnchor	0.0082	0.0023	0.0144	0.0047	0.0039	0.0014

4.2 Experimental Settings

4.2.1 Implementation Details. We encode items with frozen DistilBERT-base-uncased representations ($d = 768$) and use two-block residual adapters with bottleneck dimension 64 and dropout 0.1. All adapter methods are optimized with AdamW using a learning rate of 10^{-3} and weight decay of 10^{-4}. Each BPR batch contains 32 positives and three negatives per positive. Cross-domain experiments use five clients and train for 12×60 rounds/local steps on Music and Movies and 8×30 on Books. FedMF uses 64-dimensional factors and 25×80 rounds/local steps. FedPASA sets $\mu = 0$ during federation and performs $K_p = 40$ personalization steps with $\mu_p = .15$. Validation on Music→Movies selects $(\alpha, \kappa) = (.85, .5)$ and

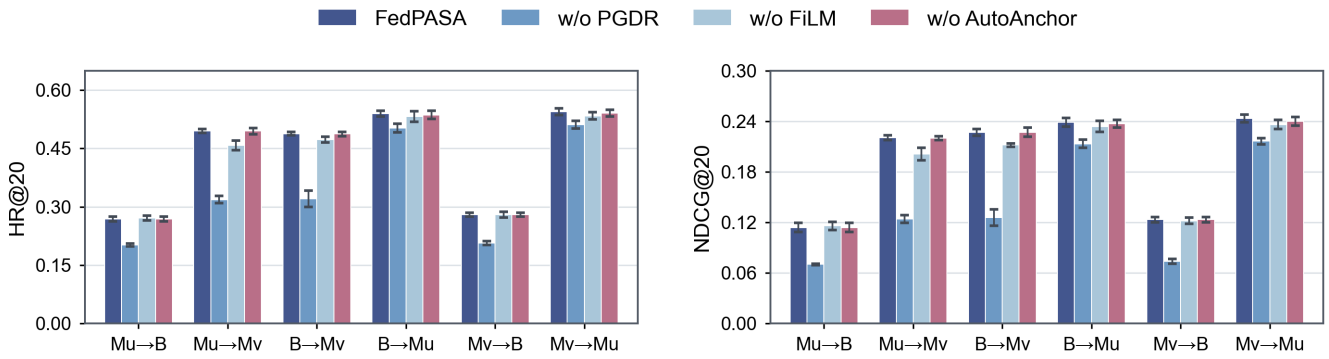


Figure 3: Component ablations under 99-negative ranking before personalization (five-seed means ± SD). Mu: Music; B: Books; Mv: Movies. PGDR contributes most; FiLM and AutoAnchor have target-dependent effects.

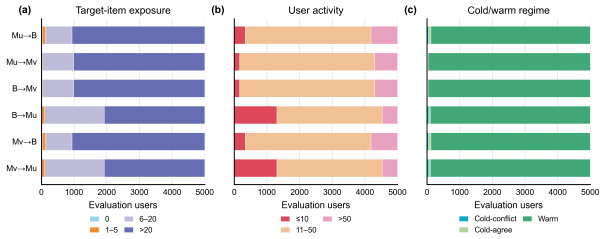


Figure 4: Evaluation-user composition: (a) target-item frequency, (b) user activity, and (c) cold/warm groups. Each transfer has 5,000 users. Mu/B/Mv: Music/Books/Movies.

$(\lambda, \rho) = (.2, .3)$, which are fixed for all targets. We report HR and NDCG. Cross-domain evaluation uses a fixed 5,000-user subset with both 99-negative and full-catalog ranking; training and validation items are excluded from candidate sets. Common-backbone results are averaged over five runs. Additional baseline settings are given in Appendix B.

4.2.2 Baselines. We compare FedPASA with representative recent and classical federated recommendation baselines. **FeDecider** [8] uses LLM representations and client-specific directional weights to personalize LoRA aggregation. **TransFR** [29] transfers a source-initialized language-model adapter and performs local personalization. **FFMSR** [16] transfers text-derived semantic representations in federated recommendation. **FedDCSR** [30] disentangles shared and domain-specific factors for federated sequential recommendation. **FedMoE** [20] personalizes expert subsets through submodel search and modular aggregation. **FedAdam** [23] applies adaptive server-side optimization, while **FedProx** [11] adds a proximal term to constrain heterogeneous local updates. **pFedMe** [5] couples personalized client parameters to a shared model through a proximal objective. **FedPer** [2] combines a globally shared representation with client-specific personalization layers. **L2-SP** [12] regularizes target parameters toward their source initialization. **FedAvg** learns one shared target adapter by averaging client updates, and **FedMF** keeps user factors local while aggregating item factors on the server. In the common-backbone comparison, all methods use the same frozen DistilBERT item features [25], user partitions, and data splits; other baselines are evaluated with their corresponding datasets and ranking cutoffs.

4.3 Main Results and Discussion (RQ1)

Evidence-dependent adaptation improves transfer across disjoint catalogs. FedPASA leads both sampled-ranking metrics on all six transfers (Tables 2 and 3), with a mean HR improvement of 14.2% over FeDecider. The margins vary by target: gains are larger for Books and Movies, whereas the baselines are already competitive on Music. Frozen text features provide a shared semantic space, but do not determine how uneven target supervision should update that space. Frequency-dependent paths offer a plausible explanation for the additional gains; the ablations below isolate the contribution of each component.

The advantage extends beyond sampled candidates. FedPASA also leads the full-catalog comparison (Table 6). The lower absolute accuracy is expected when many more items compete for the same top-ranked positions. Retaining the advantage supports improved discrimination beyond the sampled negatives, although Books remains difficult. This motivates examining both component effects and the item groups contributing to aggregate accuracy.

4.4 Ablation Study (RQ2)

Figure 3 compares FedPASA with variants that remove one component at a time. All variants are evaluated before local personalization to isolate the adaptation modules; Tables 2 and 3 report the final personalized model. Error bars show variation across five seeds.

(1) w/o PGDR. Using $h_i = A_h(x_i)$ removes frequency-dependent routing; both metrics decrease on every transfer. The equal-capacity controls retain both paths and the parameter count: constant routing removes item dependence, while shuffling breaks the frequency–weight correspondence. PGDR outperforms both controls, supporting frequency-dependent allocation of adaptation capacity (Table 10).

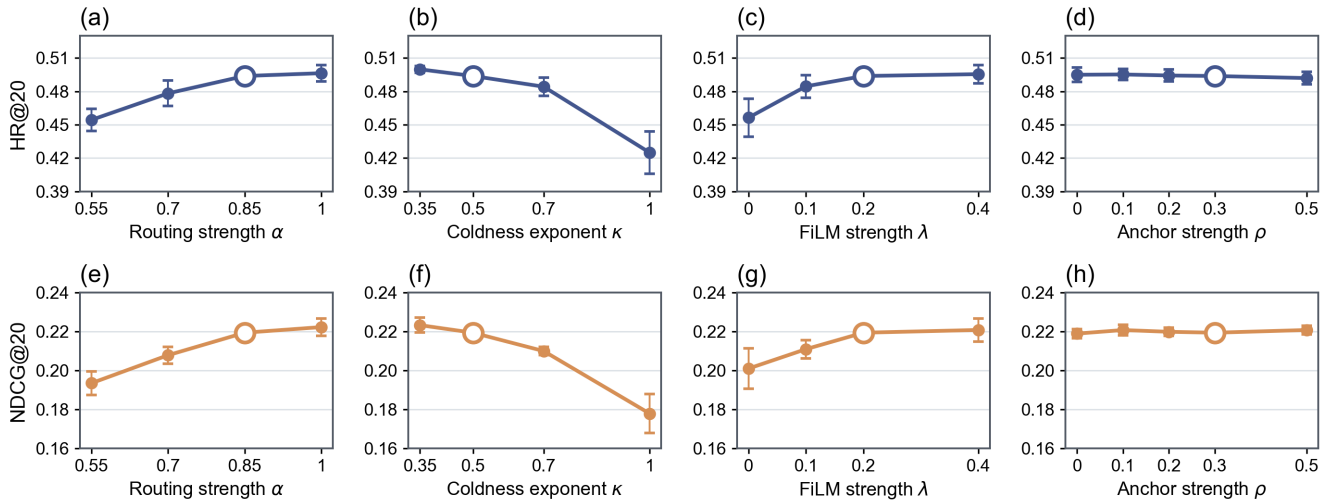


Figure 5: Parameter sensitivity on Music→Movies before personalization (99 negatives; three-seed means ± SD). Top: HR@20; bottom: NDCG@20. Each sweep varies one parameter. Open circles mark validation-selected defaults; vertical scales are shared within each row.

(2) w/o FiLM. Setting $\lambda = 0$ yields $m_{c,i} = h_i$ and removes client-dependent modulation. Average accuracy decreases, with clearer gains on Movies and Music and near-tied results on Books. Client conditioning thus complements routing, with a target-dependent contribution.

(3) w/o AutoAnchor. Setting $\rho_{\text{eff}} = 0$ removes the frozen-prior contribution in Eq. (8). Its average effect is smaller than that of routing or conditioning: Books results are identical because the catalog gate disables anchoring, while Music and Movies show modest changes. Representation drift is examined separately below.

4.5 Evidence Mechanism Analysis (RQ3)

Table 4 compares the component ablations across item groups defined jointly by frequency and local exposure. Each removal lowers both ranking metrics relative to FedPASA base, including for locally seen and locally unseen low-frequency items.

Group improvements and aggregate gains are distinct. Figure 4 shows that frequent items and warm cases dominate evaluation, while user activity and exposure vary across target catalogs. Consequently, aggregate metrics weight the groups unequally: strong relative gains in a small low-frequency group need not dominate overall accuracy. The finer slices in Table 14 show that higher-frequency items contribute most of the full-ranking gain over FedAvg. Similar source-free and source-initialized averages (Table 10) further point to target-side adaptation as the main explanation.

4.5.1 AutoAnchor and representation drift. Preservation is measured separately from ranking through cosine distance:

$$D = \frac{1}{|C||\mathcal{I}|} \sum_{c \in C} \sum_{i \in \mathcal{I}} \left(1 - \cos(z_{c,i}^{\text{anchor}}, x_i)\right). \tag{12}$$

Anchoring slightly reduces cold-item distance but increases the all-item average after training (Table 20). Although Eq. (8) moves a fixed representation toward its prior before normalization, training changes the representation itself. The observed drift and ranking effects therefore depend on the item group and optimization outcome.

4.6 Generalization and Robustness (RQ4)

FedPASA remains effective under heterogeneous conditions. FedPASA exceeds TransFR in both metrics under all three non-IID Music→Movies partitions (Table 18). More concentrated activity makes prediction harder, but the advantage persists. The table also examines client counts, fixed computation, partial participation, and update noise.

The gains are not tied to one feature encoder. The results extend to all evaluated directions with LLaMA features (Table 18), supporting semantic adaptation beyond a particular encoder.

4.6.1 Sensitivity and statistical analysis. Figure 5 separates the four adaptation controls. Increasing routing strength α improves both metrics over the tested range, whereas increasing coldness exponent κ reduces them. FiLM gains largely saturate near $\lambda = .2$. The flatter response to ρ is consistent with the catalog gate attenuating anchoring on this target. Open circles indicate defaults selected on validation data, not maxima of the plotted test curves. Table 17 gives the numerical values.

The positive confidence interval for paired HR improvements over the learned-gate baseline supports a repeatable aggregate advantage; Tables 7 and 8 report variability.

5 Conclusion

FedPASA combines routing by item frequency, client conditioning, and catalog-level anchoring for federated semantic adaptation. Experiments show cross-domain gains with a shared encoder and identify routing as the main contributor at equal model capacity. Conditioning and anchoring have group-dependent effects, motivating more selective adaptation.

References

[1] Muhammad Ammad-Ud-Din, Elena Ivannikova, Suleiman A Khan, Quang Oyomno, Qiang Fu, Kuan Ee Tan, and Adrian Flanagan. 2019. Federated Collaborative Filtering for Privacy-Preserving Personalized Recommendation System. *arXiv preprint arXiv:1901.09888* (2019).

[2] Manoj Ghuhan Arivazhagan, Vinay Aggarwal, Aaditya Kumar Singh, and Sunav Choudhary. 2019. Federated Learning with Personalization Layers. *arXiv preprint arXiv:1912.00818* (2019).

[3] Jiangxia Cao, Xixun Lin, Xin Cong, Jing Ya, Tingwen Liu, and Bin Wang. 2022. DisenCDR: Learning Disentangled Representations for Cross-Domain Recommendation. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 267–277.

[4] Zhihong Chen, Rong Xiao, Chenliang Li, Gang Ye, Hao Sun, and Hongbo Deng. 2020. ESAM: Discriminative Domain Adaptation with Non-Displayed Items to Improve Long-Tail Performance. In *Proceedings of SIGIR*.

[5] Canh T. Dinh, Nguyen H. Tran, and Tuan Dung Nguyen. 2020. Personalized Federated Learning with Moreau Envelopes. In *Advances in Neural Information Processing Systems*, Vol. 33. 21394–21405.

[6] Lei Guo, Ziang Lu, Junliang Yu, Nguyen Quoc Viet Hung, and Hongzhi Yin. 2024. Prompt-Enhanced Federated Content Representation Learning for Cross-Domain Recommendation. *arXiv preprint arXiv:2401.14678* (2024).

[7] Lei Guo, Ting Yang, Hui Liu, Xu Yu, Xiaohui Han, and Xinhua Wang. 2026. FedCRF: A Federated Cross-Domain Recommendation Method with Semantic-Driven Deep Knowledge Fusion. *Information Processing & Management* (2026). arXiv:2604.17681 doi:10.1016/j.ipm.2026.104827

[8] Xinrui He, Ting-Wei Li, Tianxin Wei, Xuying Ning, Xinyu He, Wenxuan Bao, Hanghang Tong, and Jingrui He. 2026. FeDecider: An LLM-Based Framework for Federated Cross-Domain Recommendation. In *Proceedings of the ACM Web Conference*. doi:10.1145/3774904.3792294

[9] Hoyeop Lee, Jinbae Im, Seongwon Jang, Hyunsouk Cho, and Sehee Chung. 2019. MeLU: Meta-Learned User Preference Estimator for Cold-Start Recommendation. In *Proceedings of KDD*.

[10] Tian Li, Shengyuan Hu, Ahmad Beirami, and Virginia Smith. 2021. Ditto: Fair and Robust Federated Learning Through Personalization. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*. 6357–6368.

[11] Tian Li, Anit Kumar Sahu, Manzil Zaheer, Maziar Sanjabi, Ameet Talwalkar, and Virginia Smith. 2020. Federated Optimization in Heterogeneous Networks. *Proceedings of Machine Learning and Systems* 2 (2020), 429–450.

[12] Xuhong Li, Yves Grandvalet, and Franck Davoine. 2018. Explicit Inductive Bias for Transfer Learning with Convolutional Networks. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*. 2825–2834.

[13] Xingjian Li, Haoyi Xiong, Haozhe Wang, Yifan Rao, Liping Liu, Jie Chen, and Jun Huan. 2019. DELTA: Deep Learning Transfer using Feature Map with Attention for Convolutional Networks. In *Proceedings of the 7th International Conference on Learning Representations (ICLR)*.

[14] Zhiwei Li, Guodong Long, Jing Jiang, Chengqi Zhang, and Qiang Yang. 2026. Federated Vision-Language-Recommendation with Personalized Fusion. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 40. 23337–23345. doi:10.1609/aaai.v40i28.39503

[15] Shuxia Liu, Shuang Xu, Wei Yu, Zhiqian Fu, Yongfeng Zhang, and Amelie Marian. 2021. FedCT: Federated Collaborative Transfer for Recommendation. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 716–725.

[16] Ziang Lu, Lei Guo, Xu Yu, Zhiyong Cheng, Xiaohui Han, and Lei Zhu. 2025. Federated Semantic Learning for Privacy-Preserving Cross-Domain Recommendation. *ACM Transactions on Information Systems* (2025). arXiv:2503.23026

[17] Jiaqi Ma, Zhe Zhao, Xinyang Yi, Jilin Chen, Lichan Hong, and Ed H. Chi. 2018. Modeling Task Relationships in Multi-Task Learning with Multi-Gate Mixture-of-Experts. In *Proceedings of KDD*.

[18] Tong Man, Huawei Shen, Xiaolong Jin, and Xueqi Cheng. 2017. Cross-Domain Recommendation: An Embedding and Mapping Approach. In *Proceedings of the 26th International Joint Conference on Artificial Intelligence (IJCAI)*. 2464–2470.

[19] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Aguera y Arcas. 2017. Communication-Efficient Learning of Deep Networks from Decentralized Data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*. 1273–1282.

[20] Hanzi Mei, Dongqi Cai, Ao Zhou, Shangguang Wang, and Mengwei Xu. 2024. FedMoE: Personalized Federated Learning via Heterogeneous Mixture of Experts. *arXiv preprint arXiv:2408.11304* (2024). doi:10.48550/arXiv.2408.11304

[21] Jianmo Ni, Jiacheng Li, and Julian McAuley. 2019. Justifying Recommendations Using Distantly-Labeled Reviews and Fine-Grained Aspects. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. 188–197.

[22] Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. 2018. FiLM: Visual Reasoning with a General Conditioning Layer. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 32.

[23] Sashank J Reddi, Zachary Charles, Manzil Zaheer, Zachary Garrett, Keith Rush, Jakub Konecný, Sanjiv Kumar, and H Brendan McMahan. 2021. Adaptive Federated Optimization. In *Proceedings of the 9th International Conference on Learning Representations (ICLR)*.

[24] Steffen Rendle, Christoph Freudenthaler, Zeno Gantner, and Lars Schmidt-Thieme. 2009. BPR: Bayesian Personalized Ranking from Implicit Feedback. In *Proceedings of the Twenty-Fifth Conference on Uncertainty in Artificial Intelligence (UAI)*. 452–461.

[25] Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2019. DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter. *arXiv preprint arXiv:1910.01108* (2019).

[26] Ajit Paul Singh and Geoffrey J Gordon. 2008. Relational Learning via Collective Matrix Factorization. In *Proceedings of the 14th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 650–658.

[27] Chunxu Zhang, Guodong Long, Hongkuan Guo, Xiao Fang, Yang Song, Zhaojie Liu, Guorui Zhou, Zijian Zhang, Yang Liu, and Bo Yang. 2024. Federated Adaptation for Foundation Model-Based Recommendations. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI)*. 5453–5461.

[28] Chao Zhang, Guodong Long, Tianyi Zhou, Zijian Zhang, Peng Yan, and Baodi Yang. 2024. When Federated Recommendation Meets Cold-Start Problem: Separating Item Attributes and User Interactions. In *Proceedings of the ACM on Web Conference (WWW)*. 3632–3642.

[29] Honglei Zhang, Zhiwei Li, Haoxuan Li, Xin Zhou, Jie Zhang, and Yidong Li. 2026. TransFR: Transferable Federated Recommendation with Adapter Tuning on Pre-trained Language Models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 40. 28212–28220.

[30] Huili Zhang, Dongxia Zheng, Xiang Yang, Juan Feng, and Qimei Liao. 2024. FedDCSR: Federated Cross-Domain Sequential Recommendation via Disentangled Representation Learning. In *Proceedings of the 2024 SIAM International Conference on Data Mining (SDM)*. 535–543.

[31] Jingtao Zhao, Wenjie Wang, Cong Xu, See-Kiong Ng, and Tat-Seng Chua. 2025. A Federated Framework for LLM-based Recommendation. In *Findings of the Association for Computational Linguistics: NAACL 2025*. 2852–2865.

A Dataset Statistics

Table 5 counts users retained after the three-interaction filter. Their last two events form validation and test sets. AutoAnchor uses the retained-user count divided by catalog size. Tests use a fixed subset of 5,000 users in each target domain.

Table 5: Retained data after the three-interaction filter. N **includes train, validation, and test interactions; "Eval." is the test-user count.**

Dataset	Users	Items	N	Eval.
Music	12,315	10,680	339,923	5,000
Books	262,698	170,389	10,848,388	5,000
Movies	28,226	19,252	1,056,608	5,000

Ranking masks training and validation items. The 99-negative sampler and FedMF's unpaired traversal follow §4.2. DistilBERT uses normalized masked-mean features (maximum 96 tokens). The encoder check uses LLaMA-2-7B features with a fixed signed block projection to 512 dimensions.

Metric aggregation. For the held-out item's rank r_u among unmasked candidates, define $H_u = \mathbf{1}[r_u \le K]$ and $N_u = H_u/\log_2(r_u + 1)$. HR@$K$ and NDCG@K average these quantities over the evaluation users. For disjoint nonempty groups $\mathcal{G}$ partitioning that set,

$$M = \sum_{G \in \mathcal{G}} \frac{|G|}{|\mathcal{U}_{\text{eval}}|} M_G, \qquad M \in \{\text{HR@}K, \text{NDCG@}K\}.$$

Here M_G is the group mean: aggregate gains depend on group size as well as within-group improvement.

B Full Ranking and Run-to-Run Variation

Table 6: Full-ranking comparison; each cell gives HR@20/NDCG@20, complementing Table 2. FeDecider is averaged over three runs and the common-backbone methods over five runs. Bold/underline denote the best/second-best values; FedPASA includes final personalization.

Method	M→B	M→V	B→V	B→M	V→B	V→M	Mean
FedMF [1]	0.0003/0.0001	0.0022/0.0009	0.0022/0.0009	0.0024/0.0009	0.0003/0.0001	0.0024/0.0009	0.0016/0.0006
FedAvg [19]	0.0002/0.0000	0.0037/0.0015	0.0037/0.0015	0.0125/0.0043	0.0002/0.0000	0.0125/0.0043	0.0055/0.0019
FedProx [11]	0.0001/0.0000	0.0041/0.0015	0.0041/0.0015	0.0130/0.0047	0.0001/0.0000	0.0130/0.0047	0.0057/0.0021
FedAdam [23]	0.0001/0.0000	0.0033/0.0011	0.0033/0.0011	0.0110/0.0041	0.0001/0.0000	0.0110/0.0041	0.0048/0.0018
TransFR [29]	0.0001/0.0000	0.0037/0.0012	0.0030/0.0011	0.0126/0.0047	0.0000/0.0000	0.0132/0.0050	0.0054/0.0020
L2-SP [12]	0.0000/0.0000	0.0029/0.0011	0.0028/0.0011	0.0132/0.0046	0.0001/0.0000	0.0132/0.0049	0.0054/0.0019
FedPer [2]	0.0001/0.0000	0.0034/0.0012	0.0040/0.0014	0.0127/0.0045	0.0002/0.0001	0.0119/0.0042	0.0054/0.0019
pFedMe [5]	0.0001/0.0000	0.0029/0.0009	0.0034/0.0012	0.0125/0.0046	0.0000/0.0000	0.0132/0.0049	0.0053/0.0019
FedMoE [20]	0.0001/0.0000	0.0041/0.0015	0.0045/0.0016	0.0134/0.0050	0.0002/0.0001	0.0137/0.0052	0.0060/0.0022
FeDecider [8]	0.0001/0.0000	0.0071/0.0025	0.0055/0.0019	0.0136/0.0049	0.0001/0.0001	0.0145/0.0055	0.0068/0.0025
FedPASA	**0.0017/0.0006**	**0.0126/0.0046**	**0.0140/0.0050**	**0.0177/0.0065**	**0.0018/0.0009**	**0.0188/0.0069**	**0.0111/0.0041**

Table 7: Sampled HR@20 mean and sample SD. FeDecider uses three runs; other methods use five. Mean-column SD is computed after averaging six directions within each run.

Method	M→B	M→V	B→V	B→M	V→B	V→M	Mean
TransFR [29]	0.2058±0.0051	0.3292±0.0065	0.3330±0.0210	0.5046±0.0127	0.2001±0.0071	0.5048±0.0074	0.3462±0.0066
FedMoE [20]	0.2039±0.0032	0.3424±0.0120	0.3566±0.0077	0.5088±0.0056	0.2064±0.0059	0.5107±0.0085	0.3548±0.0042
FeDecider [8]	0.2011±0.0023	0.4278±0.0055	0.3981±0.0131	0.5355±0.0240	0.2058±0.0029	0.5380±0.0214	0.3844±0.0070
FedPASA base	0.2692±0.0063	0.4950±0.0055	0.4881±0.0039	0.5394±0.0072	0.2796±0.0053	0.5449±0.0085	0.4360±0.0044
FedPASA	0.2719±0.0068	0.4980±0.0085	0.4948±0.0053	0.5408±0.0126	0.2824±0.0052	0.5462±0.0094	0.4390±0.0058

Table 8: Sampled NDCG@20 mean and sample SD. FeDecider uses three runs; other methods use five. Mean-column SD is computed after averaging six directions within each run.

Method	M→B	M→V	B→V	B→M	V→B	V→M	Mean
TransFR [29]	0.0723±0.0032	0.1283±0.0028	0.1307±0.0090	0.2121±0.0044	0.0705±0.0030	0.2115±0.0042	0.1376±0.0027
FedMoE [20]	0.0707±0.0010	0.1355±0.0035	0.1444±0.0027	0.2143±0.0031	0.0731±0.0024	0.2141±0.0059	0.1420±0.0019
FeDecider [8]	0.0714±0.0014	0.1823±0.0055	0.1705±0.0093	0.2283±0.0099	0.0734±0.0027	0.2284±0.0095	0.1590±0.0042
FedPASA base	0.1140±0.0053	0.2204±0.0031	0.2268±0.0042	0.2389±0.0051	0.1232±0.0034	0.2432±0.0046	0.1944±0.0032
FedPASA	0.1159±0.0054	0.2254±0.0053	0.2284±0.0032	0.2403±0.0076	0.1249±0.0035	0.2439±0.0069	0.1965±0.0035

Baseline settings. The FedMoE comparison uses two shared adapters with a learned gate, without client-specific submodel search or modular aggregation. FedDCSR reports target-domain results, so transfers to the same target share a value. FFMSR retains its corresponding evaluation setting.

C Proofs and Mechanism Analysis

The following derivations use the definitions in §3.6. They characterize the adaptation operators under stated assumptions; recommendation accuracy is evaluated separately.

C.1 Proof of Proposition 1

Write $\delta = x - z^*$ and $v = \sigma^2/r > 0$. Since $\mathbb{E}\varepsilon = 0$, the cross term vanishes:

$$\begin{aligned} R(\omega) &= \mathbb{E}\|(1-\omega)\varepsilon + \omega\delta\|^2 \\ &= (1-\omega)^2 v + \omega^2 b^2. \end{aligned} \tag{13}$$

Thus $R'(\omega) = 2[(v+b^2)\omega - v]$ and $R''(\omega) = 2(v+b^2) > 0$. The unique minimizer on $[0,1]$ is $\omega^* = v/(v+b^2)$, including $\omega^* = 1$ when $b = 0$. Moreover,

$$\frac{d\omega^*}{dr} = -\frac{\sigma^2 b^2}{(\sigma^2 + r b^2)^2} \le 0. \tag{14}$$

The inequality is strict for $b > 0$. This result concerns the unnormalized estimator and motivates decreasing prior weight as target reliability increases.

C.2 Proof of Proposition 2

Let $d_\omega = \|w_\omega\|$ and $n_\omega = (1-\omega)a + \omega$. The reverse triangle inequality gives $d_\omega \ge 1 - 2\omega > 0$. Expanding its square yields

$$d_\omega^2 = (1-\omega)^2 + \omega^2 + 2\omega(1-\omega)a.$$

Since $x^\top y_\omega = n_\omega/d_\omega$, $n'_\omega = 1 - a$, and $d'_\omega = (2\omega - 1)(1-a)/d_\omega$, differentiation gives

$$\begin{aligned} \frac{d(x^\top y_\omega)}{d\omega} &= \frac{(1-a)[d_\omega^2 - n_\omega(2\omega - 1)]}{d_\omega^3} \\ &= \frac{(1-\omega)(1-a^2)}{d_\omega^3} \ge 0. \end{aligned} \tag{15}$$

Here $|a| \le 1$. The result holds for fixed x, z; it does not order representations produced by different training trajectories.

D Additional Baseline Comparisons and Ablations

Table 9: Cross-domain recommendation on Amazon Music/Books/Movies [29]. Each cell contains HR@10/NDCG@10.

Method	M→B	M→V	B→V	Mean
CMF [26]	.1833/.0955	.1378/.0764	.1455/.0662	.1555/.0794
EMCDR [18]	.2149/.1121	.1893/.0901	.1652/.0705	.1898/.0909
DisenCDR [3]	.2243/.1182	.1988/.0923	.1912/.0821	.2048/.0975
FCF [1]	.1380/.0663	.1154/.0583	.1004/.0421	.1179/.0556
IFedRec [28]	.1981/.0834	.1810/.0673	.1703/.0497	.1831/.0668
FELLRec [31]	.2003/.0995	.1844/.0812	.1731/.0831	.1859/.0879
FedCT [15]	.2034/.1122	.1765/.0899	.1585/.0845	.1795/.0955
FedDCSR [30]	.2217/.1085	.1881/.0892	.1719/.0732	.1939/.0903
TransFR [29]	**.2335/.1133**	**.2123/.0934**	**.2089/.0893**	**.2182/.0987**

Table 10: Component and capacity controls: six-transfer means over five runs. Only the final row includes personalization.

Model	HR@20	NDCG@20
FedPASA base	0.4360	0.1944
w/o PGDR	0.3434	0.1373
w/o FiLM	0.4245	0.1867
w/o AutoAnchor	0.4346	0.1934
Constant routing	0.3481	0.1404
Random routing	0.3381	0.1350
Per-item anchor	0.4365	0.1950
Source-free base	0.4362	0.1958
FedMoE [20]	0.3548	0.1420
Personalized	0.4390	0.1965

Table 11: Notation used in FedPASA.

Symbol	Definition
x_i, e_u	Frozen unit item vector; private user vector
A_h, A_l, F_ℓ	Head/tail adapters; residual bottleneck MLP
q_i, q'_i, t, κ	Coldness scores, threshold, and exponent
$h_i, m_{c,i}, z_{c,i}$	Routed, modulated, and normalized features
p_c, g_ϕ	Private profile and shared conditioner
α, λ	Mixture and client-modulation strengths
$\eta_T, \rho, \rho_{\text{eff}}, \tau_l, \tau_h$	Coverage, anchor weights, and thresholds
θ, μ, μ_p	Shared parameters and proximal coefficients
r, b^2, σ^2	Reliability, prior bias, and noise scale

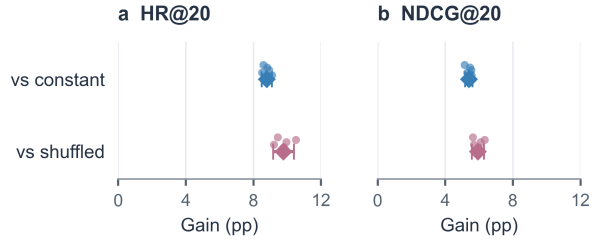


Figure 6: PGDR gains over equal-capacity routing controls (99 negatives; before personalization). Dots: six-transfer run means; diamonds/bars: five-run mean/paired 95% CI. Unit: percentage points.

Both controls retain the two adapters. Positive gains in Figure 6 isolate the benefit of matching routing weights to item frequency.

Table 12: FFMSR and PFCR recommendation performance. R: recall; N: NDCG.

FFMSR [16]

Domain	R10	R50	N10	N50
Office	.1312	.2069	.0890	.1054
Arts	.1257	.2377	.0711	.0955
OnlineRetail	.1655	.4110	.0763	.1300
Pantry	.0695	.1818	.0326	.0568

PFCR (full prompting) [6]

Domain	R10	R50	N10	N50
Office	.1206	.1938	.0782	.0941
Arts	.1153	.2199	.0677	.0904
OnlineRetail	.1561	.3982	.0739	.1266
Pantry	.0616	.1591	.0293	.0503

Table 13: FedCRF full-ranking performance across domain pairs [7].

Pair	Domain	R@10	R@20	N@10	N@20
Kitchen–Food	Kitchen	.04465	.06601	.02523	.03112
	Food	.11078	.16105	.06708	.08193
Care–Beauty	Care	.07115	.09897	.04250	.05001
	Beauty	.09342	.13384	.05398	.06513
Food–OnlineRetail	Food	.11142	.16185	.06713	.08189
	OnlineRetail	.22864	.31488	.19232	.23368

Table 14: Fine-grained frequency slices (99-negative ranking; unpersonalized models).

Frequency	Method	HR@20	NDCG@20
Unseen	TransFR [29]	0.3020	0.1070
	FedMoE [20]	0.2460	0.0927
	FedPASA base	0.5360	0.3783
1–5	TransFR [29]	0.3711	0.1562
	FedMoE [20]	0.3915	0.1662
	FedPASA base	0.3439	0.2045
6–20	TransFR [29]	0.3465	0.1395
	FedMoE [20]	0.3541	0.1438
	FedPASA base	0.2631	0.0990
>20	TransFR [29]	0.3455	0.1361
	FedMoE [20]	0.3544	0.1404
	FedPASA base	0.4956	0.2258

The unseen slice contains 20 nonempty evaluations (mean support: 13 users); other slices contain 30. Evaluations with zero hits are included.

Routing interpretation. With FiLM disabled, Eq. (7) gives the pre-normalization vector $v_i = (1-\alpha q'_i)A_h(x_i)+\alpha q'_i A_l(x_i)$. For $g_i = \nabla_{v_i}\ell$, the chain rule yields

$$\nabla_{A_h(x_i)}\ell = (1-\alpha q'_i)g_i, \qquad \nabla_{A_l(x_i)}\ell = \alpha q'_i g_i.$$

Thus shuffling routing weights changes which items train each path, even though capacity and the weight distribution are preserved.

E Hyperparameters and Robustness Analysis

Table 15: Validation-only selection on Music→Movies (mean sampled HR@20 over three runs; eight rounds, 30 local steps). Stage 1 fixes $\lambda = .4, \rho = .2$; Stage 2 fixes the selected $\alpha = .85, \kappa = .5$. The final choice is $(.85, .5, .2, .3)$.

$\alpha\backslash\kappa$	.5	.7	1.0
0.4	0.2807	0.2640	0.2517
0.55	0.3261	0.2871	0.2626
0.7	0.3657	0.3139	0.2721
0.85	0.3851	0.3541	0.2883

$\lambda\backslash\rho$	.1	.2	.3
0.2	0.3917	0.3917	0.3925
0.4	0.3879	0.3851	0.3893
0.6	0.3842	0.3874	0.3817

E.1 Configuration

Table 16: Default configuration. Sensitivity experiments vary one setting at a time.

Setting	Value
Frozen encoder	DistilBERT-base-uncased, $d = 768$
Adapter	2 residual blocks; bottleneck 64; dropout 0.1
Optimizer	AdamW; lr 10^{-3}; weight decay 10^{-4}
BPR batches	32 positives; 3 negatives/positive; logit scale 10
Federation	Cross: 12 × 60 rounds/steps; target Books: 8 × 30
Clients	5
Mixing / FiLM	$\alpha = .85$, $t = 0$, $\kappa = .5$, $\lambda = .2$
Anchor	$\rho = .3$; $[\tau_l, \tau_h] = [1, 1.5]$
Proximal terms	FedProx: .1; L2-SP: .01; base FedPASA: 0
Final personalization	40 local steps, $\mu_p = .15$ (full FedPASA and pFedMe)
FedMF baseline	64-dimensional factors; 25 rounds, 80 steps/round

Table 17: HR@20/NDCG@20 for Figure 5 (Music→Movies, three runs). (d): selected default.

Factor	Value	HR@20/NDCG@20
α	0.55	0.4543/0.1935
	0.7	0.4782/0.2079
	0.85	0.4938/0.2194 (d)
	1.0	0.4963/0.2223
κ	0.35	0.4998/0.2233
	0.5	0.4938/0.2194 (d)
	0.7	0.4842/0.2100
	1.0	0.4249/0.1779
λ	0.0	0.4563/0.2010
	0.1	0.4845/0.2110
	0.2	0.4938/0.2194 (d)
	0.4	0.4955/0.2208
ρ	0.0	0.4949/0.2190
	0.1	0.4952/0.2208
	0.2	0.4943/0.2199
	0.3	0.4938/0.2194 (d)
	0.5	0.4919/0.2208

E.2 Baseline Configuration

FFMSR uses BERT features and validation-based model selection. Pretraining/fine-tuning lasts 5/20 epochs for M–V and 1/5 epochs for pairs involving Books. FedDCSR uses 10 communication rounds with three local epochs per round. The sequential benchmarks contain 10,634 Music, 89,254 Books, and 18,643 Movies items. Tables 2 and 3 list all six evaluated transfers.

Conditioning bound. Since $\|h_i\|_2 \le 1$ and $\|\tanh \gamma_c\|_\infty \le 1$, the triangle inequality gives $\|m_{c,i} - h_i\|_2 \le \lambda(1 + \|\beta_c\|_2)$ for $\lambda \ge 0$. Thus modulation strength depends explicitly on both λ and the learned shift.

Table 18: Robustness HR@20/NDCG@20 (three-run base means). C/FC: clients/fixed compute; P: participation; LLM: LLaMA. Noise averages six transfers.

Setting	HR/NDCG	Setting	HR/NDCG
C5	0.4940/0.2183	P0.2	0.2583/0.1034
C20	0.5769/0.2788	P0.5	0.2692/0.1074
C50	0.6035/0.2935	Noise 0.1	0.4346/0.1938
Act. 0.1	0.4523/0.2059	Noise 0.2	0.4351/0.1944
Act. 0.5	0.4839/0.2167	Noise 0.3	0.4343/0.1941
Pop. 0.5	0.4895/0.2199	LLM M→V	0.5091/0.2243
FC5	0.2803/0.1089	LLM B→V	0.4869/0.2203
FC20	0.2460/0.0919	LLM M→B	0.2519/0.0998
FC50	0.2419/0.0889		

Table 19: Anchor granularity (five-run cross-domain means).

Anchoring	HR@20	NDCG@20
None	0.4346	0.1934
Catalog gate	0.4360	0.1944
Per-item gate	0.4365	0.1950

Table 20: Cosine distance from the frozen prior. Reference: TransFR.

Slice	Reference	No anchor	Base
Cross all	0.8916	0.9450	0.9841
Cross cold	0.8983	0.9721	0.9645

Fixed-compute/participation runs use eight rounds, $(\alpha, \kappa, \lambda, \rho) = (.55, .7, .4, .2)$, and 150/160/150 total steps for 5/20/50 clients. Full participation uses five clients; other rows use the selected configuration.

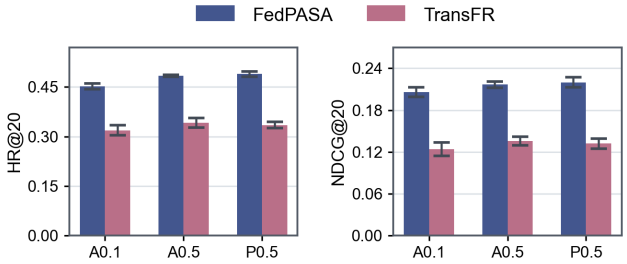


Figure 7: Non-IID Music→Movies results before personalization (99 negatives; three-run means ± SD). A/P: activity/popularity Dirichlet splits.